

BANK LOAN DATA ANALYSIS REPORT

Power BI | SQL | Python

Project Overview

The objective of this project was to analyze a bank's loan portfolio and build an interactive Power BI dashboard that helps management monitor lending performance, evaluate portfolio quality, identify borrower trends, and support data-driven lending decisions.

The project combines SQL, Python, and Power BI to transform raw loan data into meaningful business insights.

Dataset

The dataset contains information related to thousands of loan applications including:

- Loan ID
- Loan Amount
- Interest Rate
- DTI
- Purpose
- Home Ownership
- Employee Length
- Grade
- Sub Grade
- Loan Status
- Installment
- State

- Issue Date
- Total Payment
- Funded Amount

The analysis focuses on three major areas:

- Overall Loan Performance
- Customer & Loan Trends
- Loan Portfolio Quality

The dashboard enables business users to filter results by State, Grade, Purpose, and Loan Type to perform deeper analysis. The project follows the business requirements defined for Summary, Overview, and Details dashboards.

Tools Used

- Python
- Pandas
- NumPy, Seaborn, Matplotlib
- SQL Server
- Power BI
- DAX
- Excel (initial inspection)

Workflow

1. Data Cleaning

The dataset was cleaned using Python by removing duplicates, handling missing values, correcting data types, formatting dates, checking inconsistencies and preparing the dataset for SQL and Power BI.

2. SQL Analysis

SQL was used to calculate all KPIs required by the business including Total Loan Applications, Funded Amount, Amount Received, Average Interest Rate, Average DTI, Good Loan Percentage, Bad Loan Percentage, Monthly Trends, State-wise Analysis, Loan Status Analysis, Purpose Analysis and Employee Length Analysis.

3. Dashboard Development

Dashboard 1: Summary – Executive KPIs, Good vs Bad Loans, Loan Status.

Dashboard 2: Overview – Monthly Trends, State-wise Analysis, Loan Term, Employment, Purpose and Home Ownership.

Dashboard 3: Details – Complete transaction-level loan information.

KPI Analysis

Total Loan Applications: 38,576

MTD Applications: 4,314

MoM Growth: 6.9%

Total Funded Amount: \$435.8 Million

MTD: \$54 Million

MoM Growth: 13%

Total Amount Received: \$473.1 Million

MTD: \$58.1 Million

MoM Growth: 15.8%

Average Interest Rate: 12%

MTD: 12.4%

Average DTI: 13.3%

Loan Portfolio Quality

Good Loans

Applications: 33K

Percentage: 86.2%

Funded Amount: \$370.2 Million

Amount Received: \$435.8 Million

Bad Loans

Applications: 5.3K

Percentage: 13.8%

Funded Amount: \$65.5 Million

Amount Received: \$37.3 Million

Loan Status Analysis

Fully Paid: 32,145 applications, Highest amount received, Lowest average interest rate.

Charged Off: 5,333 applications, Highest credit risk, Lower repayment amount.

Current: 1,098 applications, Highest interest rate, Active repayment stage.

Monthly Trend Analysis

January: 2,332 Applications

December: 4,314 Applications

Strongest growth: September–December.

State Analysis

California dominates loan applications followed by New York, Florida and Texas.

Loan Term Analysis

36 Months: 73.2%

60 Months: 26.8%

Employment Length Analysis

Top Category: 10+ Years

Applications: Nearly 8.9K

Loan Purpose Analysis

Most Common Purpose: Debt Consolidation (~18K applications)

Other major purposes: Credit Card, Other, Home Improvement.

Home Ownership Analysis

Largest Categories: Rent, Mortgage

Smallest Categories: None, Other

Dashboard Features

Interactive filters:

- State
- Grade
- Purpose
- Good vs Bad Loan

Business Recommendations

1. Reduce Charged-Off Loans.
2. Focus on Debt Consolidation Products.
3. Expand in California, New York, Texas and Florida.
4. Reward Reliable Borrowers.
5. Monitor High-Risk Grades.

Skills Demonstrated

- Data Cleaning using Python
- Exploratory Data Analysis
- SQL Query Writing
- Data Aggregation
- KPI Design
- DAX Measures
- Power BI Dashboard Development
- Business Intelligence Reporting
- Banking Domain Analysis
- Data Storytelling
- Interactive Dashboard Design

Conclusion

This project successfully transformed raw banking loan data into an interactive business intelligence solution that enables stakeholders to monitor lending performance, evaluate portfolio quality, identify customer borrowing patterns, and support strategic decision-making. The analysis revealed a healthy loan portfolio with 38,576 loan applications, \$435.8M funded, \$473.1M recovered, and an 86.2% good loan rate, indicating strong repayment performance. Customer demand increased steadily throughout the year, with debt consolidation emerging as the primary borrowing purpose and California contributing the highest loan volume. Overall, the dashboard provides actionable insights that can help improve risk management, optimize lending strategies, and enhance operational decision-making.